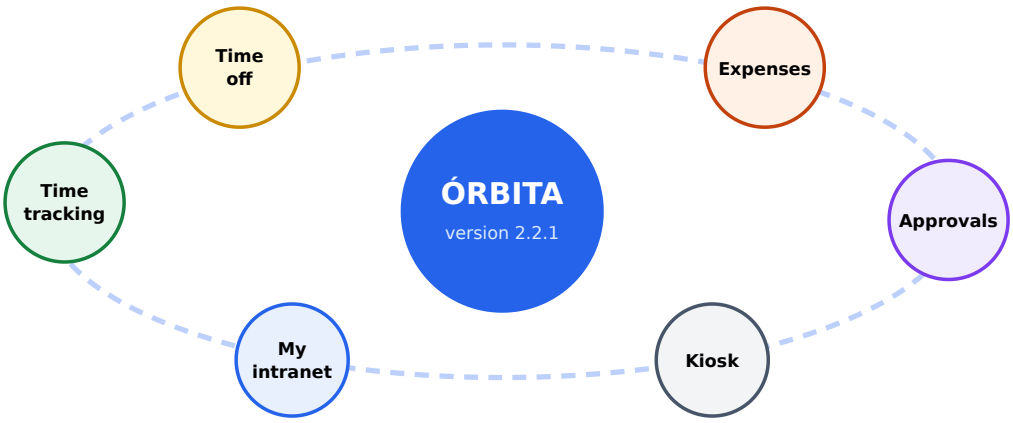
# The ÓRBITA applications, explained simply

What they are, how they are built, how they evolved from WebSphere Portal 8.5 to version 2.2.1, and how to install them with Helm, step by step.

**Repository**
`https://github.com/europesip/orbita` · chart `oci://ghcr.io/europesip/charts/hcl-dx-orbita` · images on `ghcr.io/europesip/orbita` (all public)




## Contents

1 · What ÓRBITA is
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ÓRBITA installs on an HCL DX Compose with PostgreSQL. In the labs environment ( `github.com/europesip/hcl-dx-compose-labs` ) it is Lab 5. Licence: PolyForm Noncommercial 1.0.0. Copyright © 2026 EUROPESIP.

# 1 · What ÓRBITA is

ÓRBITA is an **HR intranet** for HCL DX Compose: employees clock in and out, ask for time off and submit expenses, their managers approve them, and an **AI assistant** can do all of that in a conversation.

> **Analogy**
> If DX Compose is a shopping centre, ÓRBITA is the **staff office** set up inside it: it has a front desk in the portal (the pages), a back office where the paperwork is done (the API) and a filing cabinet (the PostgreSQL database). And since version 2.1, an **assistant** who does the paperwork for you if you ask.

## The six applications

**Time tracking** (Fichaje)
One big button: In · Out · Back. Breaks with a reason, office or remote work, location, a timeline of the day, the monthly record and corrections. Meets the Spanish working-time record law (RDL 8/2019).

**Time off** (Vacaciones)
The year's balance (22 days + 2 personal days), the whole year in a calendar where you drag to request days, and the team's schedule.

**Expenses and travel** (Gastos)
Expense reports with their receipts (photo or PDF; the text is read by OCR), a policy per category, travel requests and payments by Accounting.

**Approvals** (Aprobaciones)
The manager's inbox: time off, time-tracking corrections, expense reports and trips of their team, showing how many people are away on those days.

**My intranet** (Mi intranet)
The home page: quick clock-in, hours of the last six weeks, time off, expenses by category, what you have to approve, who is away and notifications.

**Time clock kiosk** (Quiosco)
A tablet at the entrance: each person clocks in with their PIN or by showing the QR code on their card to the camera.

## And what they all share

| Feature | What it does |
|---|---|
| **Assistant “Pregúntale a ÓRBITA”** (Ask ÓRBITA) | HCL DX IQ chat in a side panel. It acts **as the user who asks**, with 26 tools: “How many days of holiday do I have left?”, “Clock me in, working from home”, “Approve macarena's time off”. It answers in the language of the question (English or Spanish). |
| **Notifications** (bell) | Tells you when someone requests, approves, rejects, comments on or pays something of yours. |
| **Conversation** | A manager can ask (“can you send me the toll receipt?”) without rejecting the request. |
| **Receipt reading (OCR)** | When you upload the photo, shop, date, total, VAT and category are filled in. It runs in your browser (Tesseract.js): the photo is not sent to any service. |

> **Language**
> The application interface is in Spanish; the AI assistant understands and answers in English or Spanish.

# 2 · Demo users and rules

Since version 2.0 nobody has to be added to the directory: the demo story is written by **roles**, and each role is played by a **DX user that already exists**.

> **Analogy: a play**
> The script (the demo data) has characters: the HR person, the sales manager, a salesperson… The chart only needs to know **which actor** (which DX user) plays each role. That is the `demo.people` value. By default, the actors are the users of the labs OpenLDAP.



**Employee**
clocks in, requests, sees their own data

**Manager**
+ sees their team and approves its requests

**HR**
+ everybody, exports and restores the demo

**Accounting**
+ pays approved reports and exports them

| Role | User (labs) | Role in ÓRBITA | Kiosk PIN |
|---|---|---|---|
| `rrhh` | `patricia` | HR: sees everybody, exports, restores the demo | 1000 |
| `ventas-responsable` | `andres` | Sales manager: approves their team | 2001 |
| `ventas-1` | `macarena` | Sales | 1111 |
| `ventas-2` | `eduardo` | Sales | 2222 |
| `tecnologia-responsable` | `enrique` | IT manager | 3001 |
| `tecnologia-1` | `jordi` | Developer | 3333 |
| `administracion` | `german` | Accounting: pays the expense reports | 6001 |
| `tecnologia-2` , `-3` | (fictitious) | In the data; nobody signs in with them | 4444, 5555 |

> **Passwords**
> Those of these users in your directory (in the labs, the Lab 2A OpenLDAP). ÓRBITA stores no passwords.

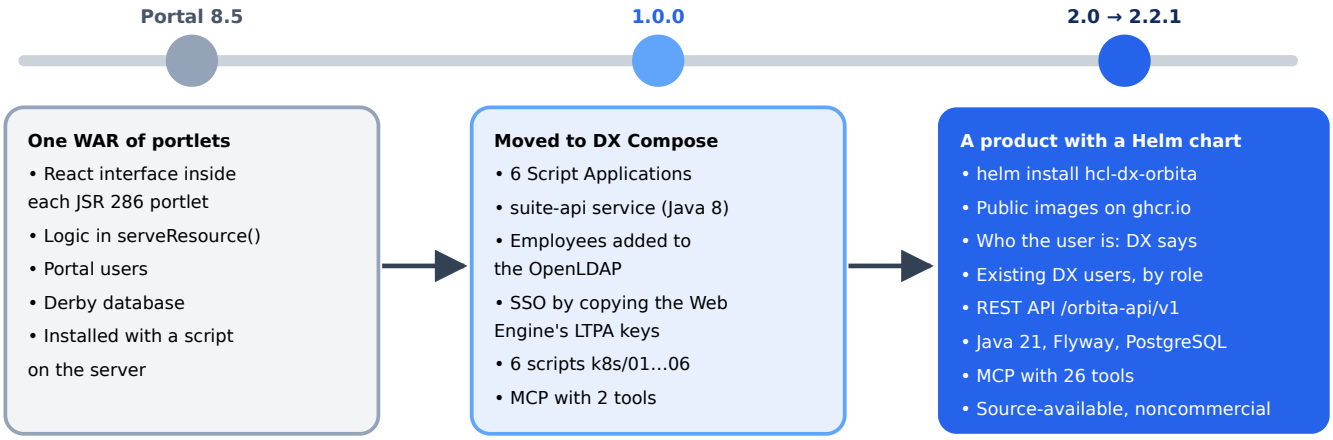
> **Restart the demo**
> HR › *Fichaje* › *Informes* › *Restaurar datos de demostración*. The data are relative to the day they are loaded.

## Business rules worth knowing

| Topic | Rule |
|---|---|
| Expected hours | Monday to Thursday 8 h, Friday 7 h (39 h a week). National holidays are loaded; HR adds local ones. |
| Clock entries | Never edited or deleted: a mistake is fixed with a **correction request** that the manager approves. |
| Breaks | Breakfast, lunch, personal errands: not working time. Doctor and client visit: paid, they count as working time. |
| Expense policy | Meals up to €30, hotel €120/night, mileage €0.26/km, other €50. Above that, a justification is required. |
| Management | People without a manager get their requests approved automatically. |

# 3 · How it was made: three stages

ÓRBITA was born as WebSphere Portal 8.5 portlets. Version 1.0 moved it to DX Compose with scripts; version 2.0 turned it into a product installed with a single Helm chart.




**Analogy**
In Portal 8.5 each application was an **all-in-one**: shop window, office and filing cabinet in the same box. Version 1.0 split it into pieces (window in the portal, office apart), but assembly was **by hand**: seven scripts and many manual steps. Version 2.0 turned it into **flat-pack furniture**: mass-produced parts (images) and a single assembly manual (the chart).

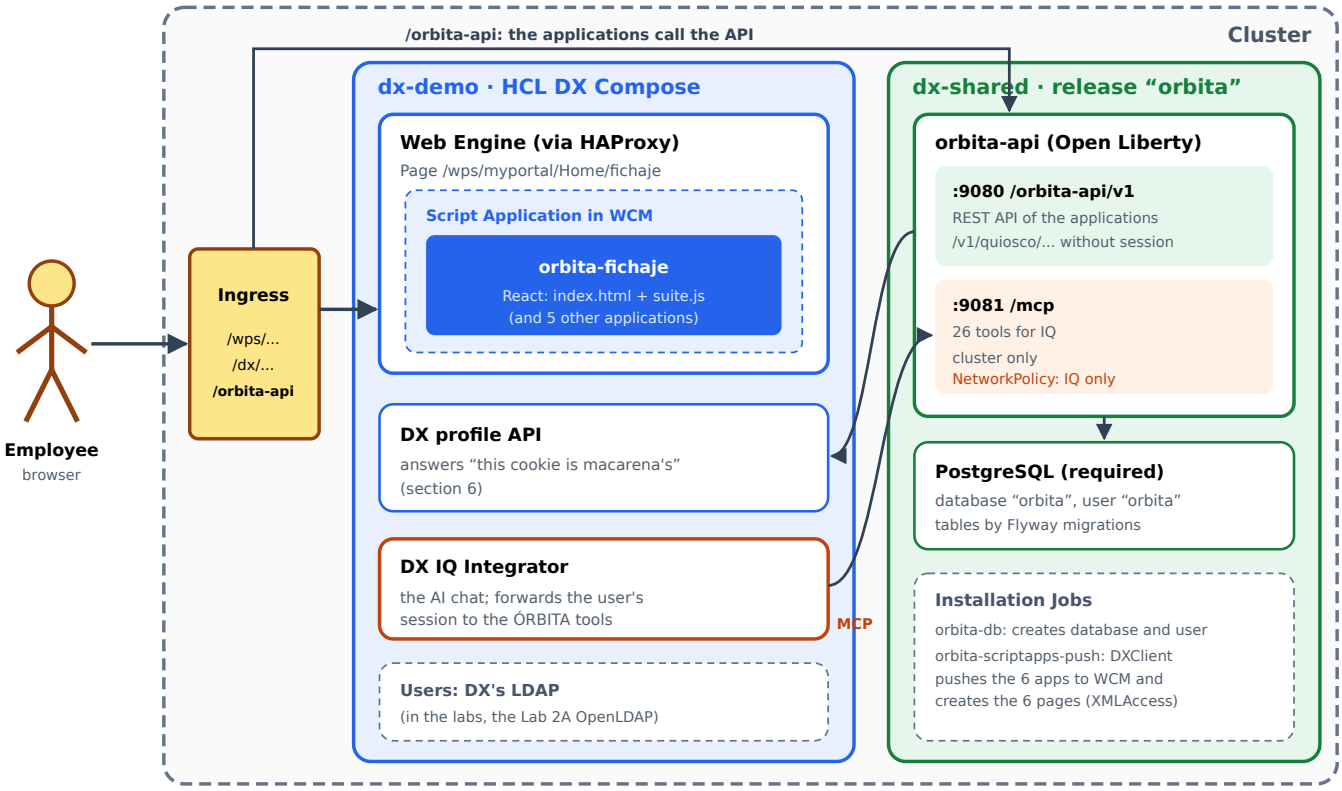
## What changed from 1.0 to 2.x

| Topic | Version 1.0 | Version 2.x |
|---|---|---|
| Installation | Scripts `k8s/01` … `06`, local build, `kubectl cp` | One chart: `helm upgrade --install` |
| Identity | Copied LTPA keys, replicated LDAP | The API asks DX who the user is |
| Demo users | `lucia.martin` … added to LDAP | Existing DX users, by role |
| Database | `suite_intranet`; a schema change dropped everything | `orbita` with Flyway migrations: data are never dropped |
| API | `POST /suite-api/api` with `op=…`, always HTTP 200 | `/orbita-api/v1` with real methods, routes and status codes |
| Code | `portlet/java/com/demo/suite`, Java 8 | `api/core\|rest\|mcp`, `com.europesip.orbita`, Java 21 |
| AI | 2 tools; the user was a parameter | 26 tools; acts as the session user |

**Portal 8.5 leftovers removed**
Version 2.0 removed the `portlet/java/com/demo/suite` folder, the `web.xml` and `ibm-web-bnd.xml` descriptors, Derby and Java 8. The React interface has been the same from the start: only the API it calls has changed.

# 4 · The map: where each piece lives

ÓRBITA is not HCL software, so it is installed in the namespace for third-party tools (`dx-shared` in the labs), next to PostgreSQL. Only its pages and Script Applications live in DX.




*The interface is served by DX; the data come from orbita-api, which asks DX who each user is; the AI (IQ) uses the MCP server in the same pod.*

**The “same domain” trick**
Everything hangs from `https://<DX host>`: the portal on `/wps`, the API on `/orbita-api` and the IQ chat on `/dx/api/iq`. So the browser sends the same session cookie everywhere, with no CORS issues or second logins.

# 5 · Inside: the code

Three layers, like a restaurant: the dining room (what you see), the kitchen (where things are done) and the pantry (where everything is kept). Plus a **menu** that dining room and kitchen share: `api/rutas.json`.




## A real REST API

`api/rutas.json` gives, for each operation of the logic, its HTTP method and route. Both the server and the interface read it, so they never get out of sync:

```
GET    /orbita-api/v1/vacaciones/resumen          → this year's balance
POST   /orbita-api/v1/vacaciones/solicitudes      {"tipo":"VACACIONES","desde":"2026-12-
09","hasta":"2026-12-11"}
DELETE /orbita-api/v1/vacaciones/solicitudes/42
# 401 without DX session · 422 {"error":"…","aviso":true} for business errors · 404 · 405 · 500
```

| Area | Routes (under `/orbita-api/v1/`) |
|---|---|
| General | `yo`, `inicio`, `config`, `salud` (health) |
| Time tracking | `fichaje/hoy`, `fichaje/marcas`, `fichaje/meses/{year}/{month}`, `fichaje/correcciones`, `fichaje/equipo`, `fichaje/informes/…` |
| Time off | `vacaciones/resumen`, `vacaciones/solicitudes`, `vacaciones/calendario/…`, `vacaciones/anios/{year}` |
| Expenses and travel | `gastos`, `gastos/notas/{id}`, `gastos/lineas`, `gastos/pagos`, `viajes` … |
| Approvals | `aprobaciones`, `aprobaciones/{class}/{id}`, `comentarios/{class}/{object}`, `avisos` |
| Kiosk (public) | `quiosco/identificacion`, `quiosco/marcas`: no session, protected by the device key |

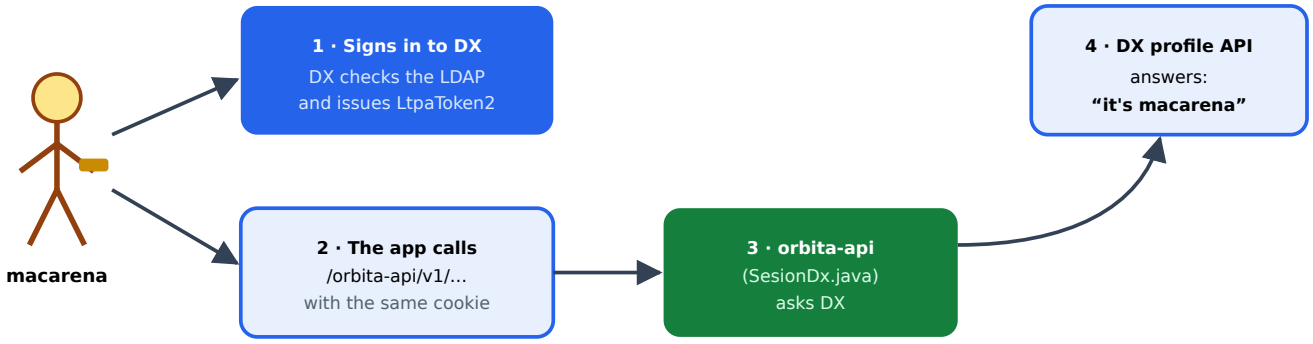
**One single brain**
The REST API and the MCP server call the **same** function, `Despachador.ejecutar(op, parameters, user)`. That is why the AI has exactly the same permissions as the user on screen.

**The schema evolves without losing data**
Tables are created and upgraded by **Flyway migrations** (`V1__esquema.sql`, `V2__…`) when `orbita-api` starts. A published migration is never edited: a new one is added.

# 6 · Who the user is: DX says

The employee signs in to the portal once and ÓRBITA already knows who they are. Since version 2.0, ÓRBITA does not work it out on its own: **it asks DX**.

> **Analogy: the festival wristband**
> When you enter the portal you get a **wristband** (the `LtpaToken2` cookie). In version 1.0, ÓRBITA kept a **copy of the secret stamp** to check wristbands itself (copied LTPA keys). In 2.0, when you arrive with your wristband, ÓRBITA **phones the box office** (the DX profile API) and asks “whose wristband is this?”. It keeps no stamps, no guest lists and nobody's keys.



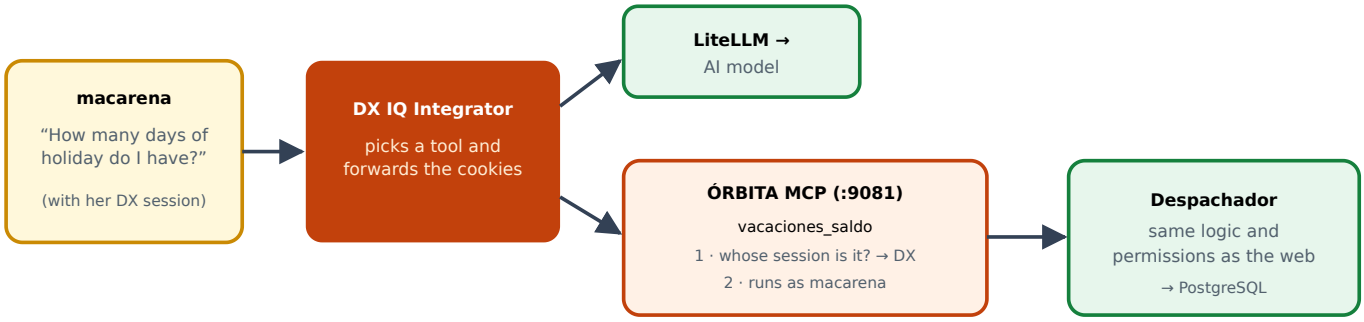

| Advantage | Why it matters |
|---|---|
| No copied LTPA keys | Version 1.0 copied the Web Engine's (the same in every CF238 installation: not fit for production). They are no longer needed. |
| No LDAP configuration | ÓRBITA does not know the directory: anything DX uses works (LDAP, Active Directory…). |
| No stored credentials | The API stores no DX administrator or LDAP bind passwords. |
| The AI cannot impersonate | The MCP server asks the same question with the session that IQ forwards (section 7). |

> **The DX certificate**
> To ask over HTTPS, the API must trust DX's certificate. If it is self-signed or from a private CA (as in the labs), you give it the CA in a Secret ( `dx.caSecret` ). `dx.insecureSkipTLSVerify` exists for labs only.

> **What about the kiosk?**
> Its page requires a session (the entrance tablet signs in with a user). But its part of the API ( `/v1/quiosco/…` ) is public: it identifies each person by PIN or QR code, and the device by a key the chart generates (Secret `orbita-kiosk` ).

# 7 · The AI: the MCP server

The “Pregúntale a ÓRBITA” button (or the IQ button in DX) opens a chat with HCL DX IQ. To talk about time off or expenses, ÓRBITA lends it a **toolbox with 26 tools**: its MCP server.




## The 26 tools, by area

| Area | Tools | ≈ tokens |
|---|---|---|
| `general` | quien_soy, resumen_dia, avisos, avisos_marcar_leidos | 470 |
| `fichaje` | fichaje_hoy, fichaje_mes, equipo_ahora, **fichar** (clock in/out), fichaje_corregir | 960 |
| `vacaciones` | vacaciones_saldo, vacaciones_solicitudes, ausencias_equipo, vacaciones_solicitar, vacaciones_cancelar | 910 |
| `gastos` | gastos_resumen, nota_gasto, nota_gasto_crear, gasto_anadir, nota_gasto_enviar, viaje_solicitar, viaje_cancelar, pagos_pendientes | 1,550 |
| `aprobaciones` | aprobaciones_pendientes, **solicitud_resolver** (approve/reject), solicitud_comentarios, solicitud_comentar | 780 |

**Every tool costs tokens**
Its description travels in **every** call to the model, used or not. All 26 are about 4,700 tokens (HCL's own, about 30,000). With `mcp.tools` you publish only some areas ( `enabled` ), remove others ( `disabled` ) or only the 14 query tools ( `readOnly: true` ).

**Left out on purpose**
Paying, restoring the demo, deleting and exporting are not AI tools: they are bulk or irreversible actions that should not be one sentence away.

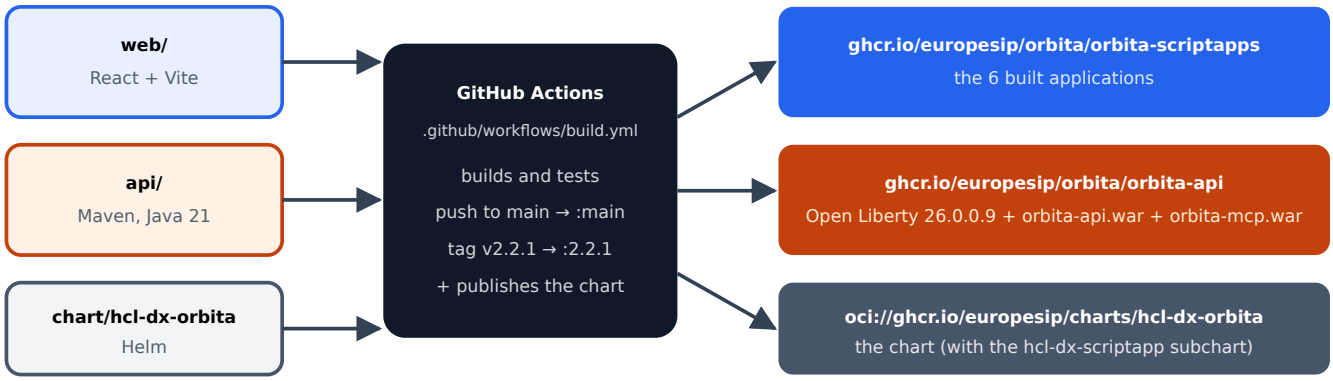
## Security

| Layer | How |
|---|---|
| **Identity** | IQ forwards the chat user's DX cookies and ÓRBITA asks DX who it is. **There is no “user” parameter**: the model cannot choose on whose behalf it acts, and no malicious text (prompt injection) can make it impersonate someone else. |
| Network | Port 9081 is not on the Ingress; a NetworkPolicy only lets the Integrator in. |
| Confirmation | Anything that changes data first returns a summary and only acts with `confirmar: true` . |
| Permissions and audit | Every operation checks what that person can do, and every call writes `ORBITA MCP auditoria herramienta=… usuario=…` to the log. Optional Bearer token ( `mcp.tokenSecret` ). |

**Two ways of adding AI to ÓRBITA**
**1.** With the HCL DX IQ Integrator (what ÓRBITA does today): the model configured in IQ and all its tools at once, DX's (WCM, DAM, pages) and ÓRBITA's. **2.** With your own chat: the MCP server is standard and the API is described in `rutas.json` ; the only condition is forwarding the DX session (or an OAuth identity exchange if the chat lives outside DX).

# 8 · How it is built and published

In 1.0 you compiled on your own machine. Now **GitHub Actions** does it: it builds, tests and publishes two images and the chart, all public.




## The repository

```
api/                  back end (Maven, Java 21, Jakarta EE 10)
  rutas.json          REST API contract: method + route → operation
  core/               logic (com.europesip.orbita.core): operations, permissions, Flyway, demo, DX
session
  rest/               orbita-api.war: REST API (/orbita-api/v1)
  mcp/                orbita-mcp.war: MCP server
web/                  React interface (Vite); scripts/scriptapps.mjs generates the 6 Script
Applications
images/               Dockerfiles: orbita-api (Open Liberty) and orbita-scriptapps
chart/hcl-dx-orbita/  Helm chart
scripts/              helpers for the labs: install, verify, register the MCP server in IQ
docs/                 specification, MCP, AI questions, REST Client tests
```

| To build locally | Command |
|---|---|
| Back end | `mvn -f api/pom.xml package` (Java 21) |
| Interface | `cd web && npm ci && npm run build:scriptapps` |

**The WCM "[$" trap**

WCM mistakes `[$…]` for one of its own tags and **removes** it when serving the JavaScript, which breaks the code ( `[$,H]=useState()` ). `web/scripts/scriptapps.mjs` writes it with equivalent escapes.

**Another quirk: Flyway on Open Liberty**

Flyway only finds the migrations if their folder contains a `flyway.location` file. That is why there is one in `db/migration` .

# 9 · Tutorial · install with Helm

Two secrets, a values file and one command

## Before you start

| You need | Details |
|---|---|
| **HCL DX Compose** | CF239 on Kubernetes with an Ingress (ingress-nginx) for its host name, and a DX **administrator** user for DXClient. |
| **PostgreSQL** | **Required in 2.x** (tested with PostgreSQL 18 / CloudNativePG), plus a Secret with a **superuser** ( `username` , `password` ). In the labs: `external-deps/02-postgres.sh` (~3 min) or Lab 2B. |
| **Users** | Those in `demo.people` (in the labs, the Lab 2A OpenLDAP users). |
| **HCL DX IQ** (optional) | For the assistant (Lab 3 of the labs). |
| **Tools** | `kubectl` and Helm 3.8 or later. Images and chart are public: no login. |

**1** **Secrets** (once): the DX administrator for DXClient and the CA of the DX certificate.

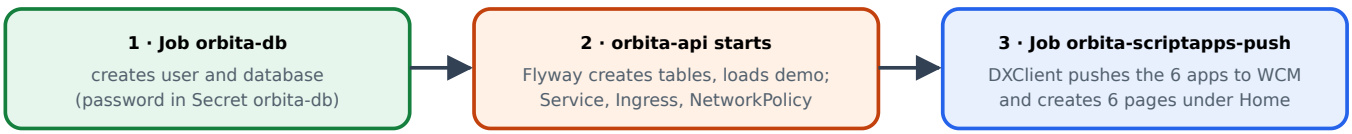
```
NS=dx-shared
kubectl -n $NS create secret generic orbita-dx --from-literal=username=wpsadmin --from-
literal=password='<password>'
kubectl -n dx-demo get secret dx-tls-cert -o jsonpath='{.data.tls\.crt}' | base64 -d > dx-ca.crt
kubectl -n $NS create secret generic orbita-dx-ca --from-file=ca.crt=dx-ca.crt
```

**2** **Values** ( `orbita-values.yaml` ):

```
dx:         { host: dx-demo.apps.promox.europesip-lab.com, caSecret: orbita-dx-ca }
database:   { host: dx-pg-rw.dx-shared.svc.cluster.local, create: { superuserSecret: dx-pg-
superuser } }
mcp:        { networkPolicy: { integratorNamespace: dx-demo } }
scriptapps:                                  # the subchart does not see the values above:
repeat them
  dx:       { host: dx-demo.apps.promox.europesip-lab.com, credentialsSecret: orbita-dx,
caSecret: orbita-dx-ca }
  dxclient: { acceptLicense: true }          # you accept the HCL DXClient licence
```

**3** **Install**:

```
helm upgrade --install orbita oci://ghcr.io/europesip/charts/hcl-dx-orbita --version 2.2.1 \
  -n dx-shared -f orbita-values.yaml --timeout 15m
```




**In the labs, with a script**

It reads the labs `env.sh` and creates the secrets for you:

```
DXCLIENT_ACCEPT_LICENSE=yes scripts/instalar-en-labs.sh
scripts/verificar.sh
```

# 10 · Connect to IQ, check and run the demo

## Register the MCP server in IQ (for the assistant)

The IQ Integrator must know the base URL `http://orbita-api.dx-shared.svc.cluster.local:9081`. HCL documents the `MCP_SERVER_LIST` variable for this, but it **has no effect** (HCL case CS1479102, still present in CF239 / IQ 1.0.3): the Integrator reads another list, in the `dx-iq-integrator` ConfigMap. Until HCL fixes it, that list is patched:

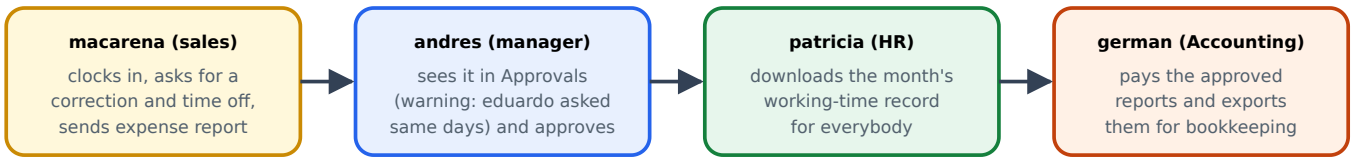
```
scripts/registrar-mcp-en-iq.sh       # in the labs: adds ÓRBITA to the list, restarts and shows:
#   Fetched 26 tools from http://orbita-api.dx-shared.svc.cluster.local:9081
```

> **After every IQ `helm upgrade`, again**
> The IQ chart regenerates the list and ÓRBITA disappears. With Helm 4 that upgrade also needs `--force-conflicts`. And after changing `mcp.tools`, restart the Integrator so it reads the new list.

## Check

| Check (`scripts/verificar.sh`) | What a failure means |
|---|---|
| Release `orbita` deployed | The `helm install` did not finish. |
| Job `orbita-db` completed | Check the PostgreSQL host and the superuser Secret. |
| `orbita-api` ready, Flyway up to date | See `kubectl -n dx-shared logs deploy/orbita-api`. |
| Job `orbita-scriptapps-push`: 6 apps in WCM | DX credentials, CA or DXClient licence. |
| `/orbita-api/v1/salud` → 200; `/v1/yo` without session → 401 | Ingress or certificate. |

## Demo walkthrough




| Area | Try in the AI chat, for example |
|---|---|
| My day | “What's on for me today?” · “Do I have anything to approve?” |
| Time tracking | “Clock me in, working from home” · “Which days this month did I not clock out?” |
| Time off | “How many days of holiday do I have left?” · “Book me holidays from 9 to 11 December” |
| Expenses | “Create a report ‘Visit to Seville’ and add yesterday's lunch, €42.80” · “Send it to my manager” |
| Approvals | As andres: “Approve macarena's time off” · “Reject the Barcelona trip: tighter budget, please” |

More than 60 example questions in `docs/ai-questions.md` (Spanish: `docs/preguntas-ia.md`).

# 11 · Upgrade, uninstall and change things

## Upgrade

```
helm upgrade orbita oci://ghcr.io/europesip/charts/hcl-dx-orbita --version <new> -n dx-shared -f
orbita-values.yaml
```

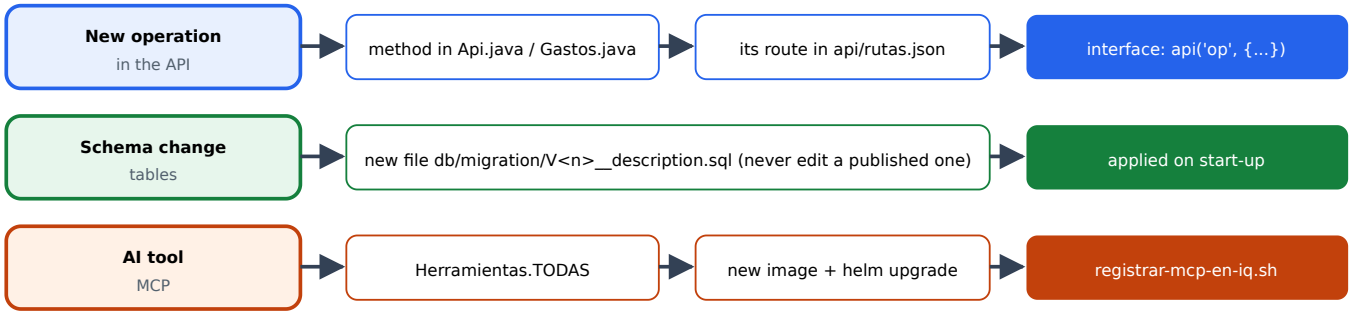
New migrations run on start-up and **data are kept**. The Script Applications are updated on their pages.

## Uninstall

```
helm -n dx-shared uninstall orbita
```

Kept on purpose: the database and its `orbita-db` Secret, the Script Applications and the pages (someone may be using them). Delete them by hand if you no longer need them, and remove ÓRBITA from the IQ MCP list with the same patch, without its URL.

## Changing things (development)




## Example: an AI tool

```
new Herramienta("fichaje_hoy", "Fichaje de hoy",
    "Indica si el usuario ha fichado hoy, desde qué hora y cuánto lleva trabajado. Úsala para '¿he
fichado?'.",
    Esquema.nuevo(), true,                      // true = query only; no user parameter
    l -> l.api("fichaje.estado")),
```

| Rule | Why |
|---|---|
| Name in `snake_case` | IQ only accepts letters, digits, `_` and `-`. |
| A good description | It is the only thing the model reads to decide whether to use it (and it travels in every request). |
| `Esquema.nuevo()` | No user parameter: it always acts as the DX user who asks. |
| If it changes data | `soloLectura = false`, `.confirmar()` and `l.pedirConfirmacion("summary")` until it is confirmed. |

**Testing the MCP server by hand**

`kubectl -n dx-shared port-forward svc/orbita-api 9081:9081` (the local port must be 9081) and `docs/mcp-pruebas.http` with a user's `LtpaToken2` cookie. For Claude Code, only in development with `ORBITA_MCP_DEV_USER` (the chart never sets it).

# 12 · Release history

Summary of the repository's `CHANGELOG.md`. All releases are dated 6 November 2026.

## 2.2.1 Ask in English or Spanish

- The “Pregúntale a ÓRBITA” panel answers in the language of each question (before, always in Spanish), and the MCP server tells the model to answer in the user's language and translate ÓRBITA's data.
- English examples in the English documentation (README, Lab 5).

## 2.2.0 Choose the MCP tools

- `mcp.tools` in the chart: publish only some areas or tools ( `enabled` ), remove some ( `disabled` ) or only the query ones ( `readOnly` ). A tool that is not published cannot be called either.
- Installation without registry login (images and chart are public), and Lab 5 in the HCL DX Compose labs.

## 2.1.0 The AI can do much more

- MCP server with **26 tools** (2 before): my day and notifications, time tracking (including clocking in and out), time off, expenses and travel (create reports, add expenses, send them) and approvals (approve, reject, comment). Everything that changes data asks for confirmation.
- More than 60 example questions ( `docs/ai-questions.md` , `docs/preguntas-ia.md` ) and the two ways of adding AI to ÓRBITA.
- **Public** repository, images and chart.

## 2.0.0 Helm chart

| Area | Changes |
|---|---|
| Installation | `hcl-dx-orbita` chart instead of the `k8s/01–06` scripts: a Job for the database, a Deployment for the API and the `hcl-dx-scriptapp` 0.2.0 subchart for apps and pages. Images `orbita-api` (Open Liberty 26.0.0.9, Java 21) and `orbita-scriptapps` built by CI. Requires PostgreSQL. |
| Users and security | Who the user is, DX says (profile API): no copied LTPA keys, no replicated LDAP, no administrator credentials. The MCP server acts as the chat user (the `usuario` argument is gone). Demo users by role ( `demo.people` ). Kiosk key generated by the chart. |
| Data | Flyway migrations: an upgrade never drops data. PostgreSQL only (Derby is gone). Database and user `orbita` . Restoring the demo runs `TRUNCATE` and reloads. |
| Code and API | Repository restructured ( `api/` , `web/` , `images/` , `chart/` , `docs/` ), package `com.europesip.orbita` . REST API `/orbita-api/v1` with real status codes, described in `api/rutas.json` . Portal 8.5 leftovers removed. Pages and kiosk key come from `GET /v1/config` . |
| Licence | PolyForm Noncommercial 1.0.0. |

## 1.0.0 First version for DX Compose

- The Portal 8.5 suite migrated to HCL DX Compose CF238 without portlets: six Script Applications and the `suite-api` service (Open Liberty 26.0.0.8).
- Installed with scripts `k8s/01-construir.sh` … `06-mcp-iq.sh` ; employees added to the OpenLDAP ( `lucia.martin` …); database `suite_intranet` ; SSO with copied LTPA keys; first MCP server with 2 tools.

# 13 · Pitfalls and cheat sheet

| Pitfall | What happens and what to do |
|---|---|
| IQ loses ÓRBITA | After every IQ `helm upgrade`: run `scripts/registrar-mcp-en-iq.sh` again (HCL bug CS1479102). |
| You change `mcp.tools` and nothing happens | The Integrator reads the tools on start-up: `kubectl -n dx-demo rollout restart deployment/dx-iq-integrator`. |
| WCM removes `[$…]` | It would break the JavaScript; `scriptapps.mjs` escapes it. |
| Self-signed DX certificate | The API cannot ask who the user is: give it the CA with `dx.caSecret`. |
| `/probe/ready` returns 503 | Normal until `orbita-api` finishes the migrations. |
| MCP tunnel on another port | The virtual host only accepts `Host: …:9081`: the local port must be 9081. |
| Testing `vacaciones_solicitar` | With `confirmar: true` it creates a real request: cancel it or restore the demo. |
| Receipt OCR | The browser needs internet the first time to download the engine and the Spanish model. |

## Cheat sheet

```
# ── Install (requirements: DX Compose, PostgreSQL + superuser Secret) ──────
kubectl -n dx-shared create secret generic orbita-dx --from-literal=username=wpsadmin --from-
literal=password='…'
kubectl -n dx-shared create secret generic orbita-dx-ca --from-file=ca.crt=dx-ca.crt
helm upgrade --install orbita oci://ghcr.io/europesip/charts/hcl-dx-orbita --version 2.2.1 \
  -n dx-shared -f orbita-values.yaml --timeout 15m

# ── In the labs ─────────────────────────────────────────────────────────────
DXCLIENT_ACCEPT_LICENSE=yes scripts/instalar-en-labs.sh
scripts/registrar-mcp-en-iq.sh        # AI assistant (optional)
scripts/verificar.sh

# ── Look ────────────────────────────────────────────────────────────────────
kubectl -n dx-shared logs job/orbita-scriptapps-push -c dxclient | tail -3
curl -k https://<DX host>/orbita-api/v1/salud                    # → ok
kubectl -n dx-shared logs deploy/orbita-api | grep "ORBITA MCP"    # tools and audit
kubectl -n dx-demo logs deploy/dx-iq-integrator | grep Fetched    # Fetched 26 tools ...

# ── Upgrade / uninstall ─────────────────────────────────────────────────────
helm upgrade orbita oci://ghcr.io/europesip/charts/hcl-dx-orbita --version <new> -n dx-shared -f orbita-
values.yaml
helm -n dx-shared uninstall orbita
```

**4 ideas to remember**

1. The interface is 6 **Script Applications** in WCM.
2. The logic is **orbita-api**, installed with **one chart**.
3. **DX says who the user is**: ÓRBITA keeps no keys.
4. The AI uses the **same logic and permissions** via MCP.

**Further reading**

`github.com/europesip/orbita`
`README.md` · `CHANGELOG.md`
`docs/mcp.md` · `docs/ai-questions.md`
`chart/hcl-dx-orbita/values.yaml`
Services: `www.europesip.com`

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